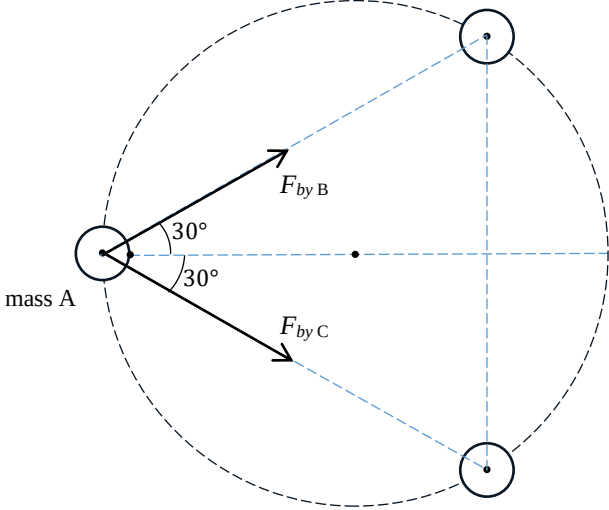


	2. $\Sigma p_y = 0 \text{ kg m s}^{-1}$	B1
(b)(ii)	By principle of conservation of linear momentum, ($\rightarrow$) $7.3 \times 10^{-22} = (p_p)(\cos 60^\circ) + (p_e)(\cos 25^\circ) \quad \dots(1)$ ($\uparrow$) $(p_e)(\sin 25^\circ) = (p_p)(\sin 60^\circ) \quad \dots(2)$ Solving (1) and (2) gives, $\frac{(p_p)(\sin 60^\circ)}{(p_p)(\cos 60^\circ)} = \tan 60^\circ = \frac{(p_e)(\sin 25^\circ)}{(7.3 \times 10^{-22}) - (p_e)(\cos 25^\circ)}$ $\Rightarrow p_e = 6.35 \times 10^{-22} \text{ kg m s}^{-1}$ M1, M1 – 2 equations showing application of COLM A1 – final answer after solving two equations.	M1 M1 A1

Q2		
(a)	Every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them .	B1 B1
(b)(i)	 The vertical components of the two forces due to B and C cancel each other, hence, Resultant force in the y-direction = 0 Hence, Resultant force F = Resultant force in the x-direction $= 2 \times \frac{GM^2}{d^2} \times \cos 30^\circ = 2 \times \left(\frac{6.67 \times 10^{-11} (6.20 \times 10^{24})^2}{(1.32 \times 10^9)^2} \right) \times \cos 30^\circ = 2.5487 \times 10^{21} \text{ N}$ $= 2.55 \times 10^{21} \text{ N}$ Alternative methods accepted.	B1 M1
(b)(ii)	The resultant gravitational force provides for the centripetal force required for the rotation of planet A. $F = m\omega^2 R$ $\omega = \sqrt{\frac{F}{mR}} = \sqrt{\frac{2.5487 \times 10^{21}}{6.20 \times 10^{24} \times 7.60 \times 10^8}} = 7.3546 \times 10^{-7} \text{ rad s}^{-1}$ $T = \frac{2\pi}{\omega} = \frac{2\pi}{7.3546 \times 10^{-7}} = 8.54 \times 10^6 \text{ s}$	B1 M1 A1
(b)(iii)	The gravitational potential is set to be zero at infinity. Gravitational force is attractive in nature and the force exerted on a test mass by the external agent will be in opposite direction to the displacement of the mass. Thus negative work is done by the external force (to bring a test mass from infinity to the point and hence potential is negative). <u>OR</u> The gravitation potential is set to be zero at infinity.	B1 B1 OR B1

	Since gravitational force is attractive in nature, (positive) work is done by an external agent to bring a (test) mass from the point (in the field of these 3 planets) to infinity , hence the (initial) potential is therefore (lower than at infinity and hence) negative.	B1
(b)(iv)	$ \begin{aligned} U &= U_{AB} + U_{AC} + U_{BC} \\ &= -\frac{Gm^2}{d} + \left(-\frac{Gm^2}{d}\right) + \left(-\frac{Gm^2}{d}\right) \\ &= 3 \times \left(-\frac{6.67 \times 10^{-11} (6.20 \times 10^{24})^2}{1.32 \times 10^9} \right) \\ &= -5.83 \times 10^{30} \text{ J} \end{aligned} $	M1 A1

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